



Sustainability

A Call to Action:

SUSTAINABILITY ANALYSIS



TEAM MEMBERS



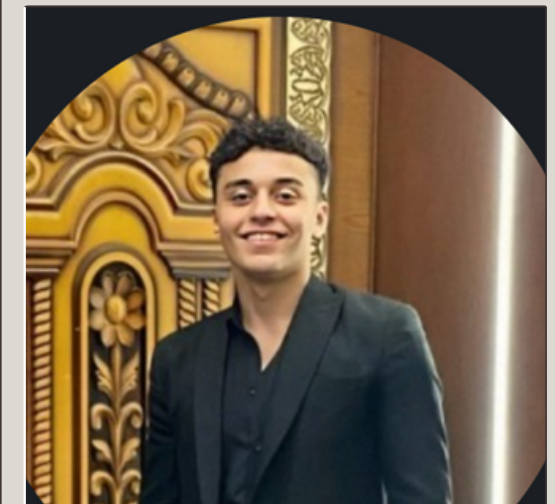
GEORGE ESMAT



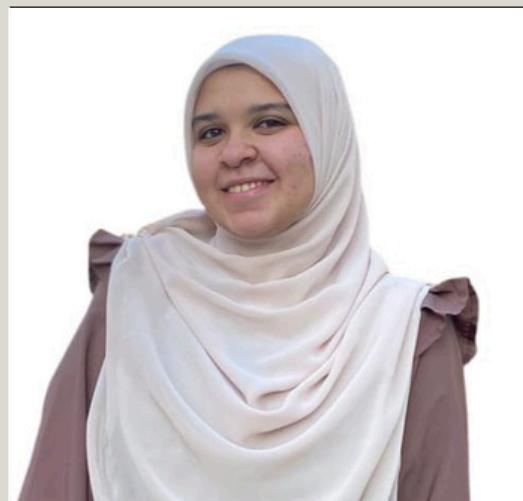
MALAK AHMED



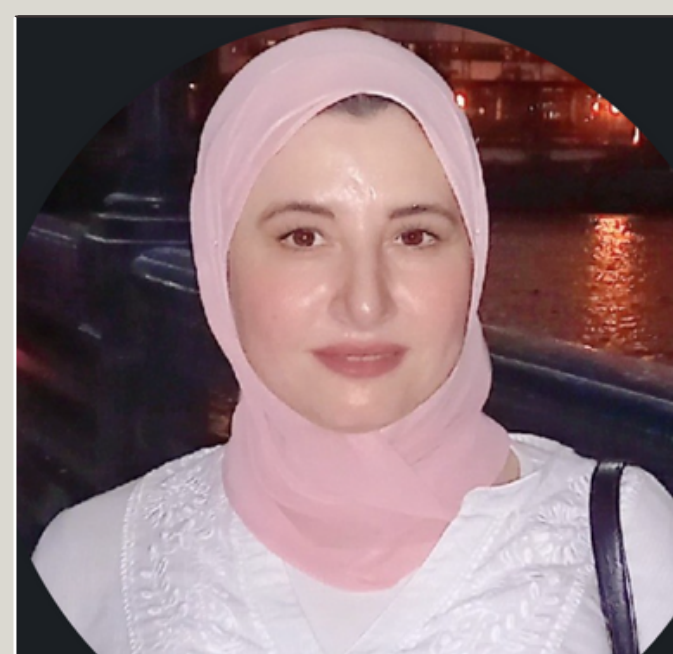
DINA ALI



NOOR MAGED



DOAA SHABBAN

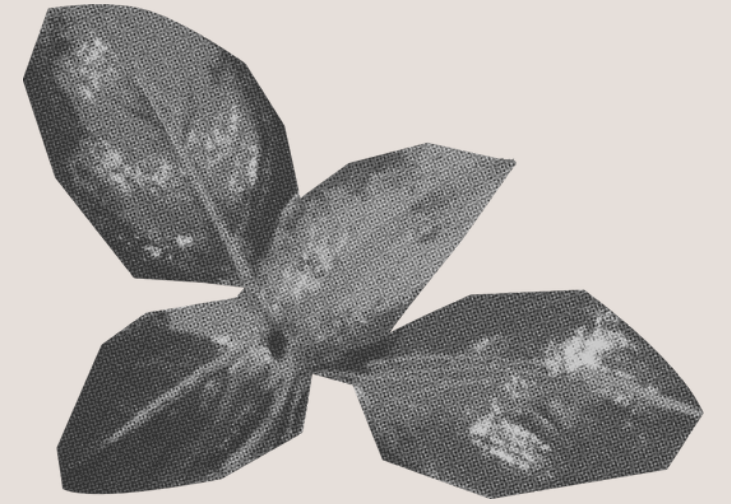


**UNDER SUPERVISION OF
DR/ AMAL MAHMOUD**



ESRAA REFAAT

AGENDA



1 Introduction

Objective

2

3 Methodology

Excel

3.1

3.2 SQL

Python

3.3

3.4 Tableau

Power BI

3.5





INTRO



In a world where the shift towards environmentally responsible practices is accelerating, sustainability is no longer a luxury but a strategic priority for every organization striving for excellence and competitiveness.

In this project, we started with raw data and ended with a comprehensive, data driven analytical perspective. We performed data cleaning to ensure accuracy and error free results, then designed a data model to simplify the analysis and linking of tables.

Next, we moved to the analysis and insights extraction phase to uncover patterns of environmental performance and their impact on the market. Finally, we created an interactive dashboard that enables monitoring of sustainability indicators and making decisions based on accurate and clear data.

rating	eco_friendly_manufacturing	carbon_footprint	water_usage	waste_production	recycling_programs	average_price	product name	sustainability_score
2.23	No	19.3	391	2.81	No	14.88	Ecotowel	Not Sustainable
2.1	No	32.74	171	9.08	No	2.21	Terrashirt	Not Sustainable
1.74	No	29.26	1385	8.21	No	3.06	Ecopack	Not Sustainable
3.08	No	4.11	2323	4.4	No	44.37	Purejar	Not Sustainable
2.83	No	115.83	566	14.81	No	24.31	Ecopack	Not Sustainable
3.3	No	39.42	1063	2.89	No	19.36	Terrashirt	Not Sustainable
2.87	No	23.43	4080	2.89	No	50.89	Greenbottle	Not Sustainable
1.8	No	21.28	36	1.51	No	24.41	Purepack	Not Sustainable
1.25	No	389.92	149	1.6	No	22.84	Ecotowel	Not Sustainable
2.43	No	37.74	1189	8.06	No	16.52	Biotoy	Not Sustainable
1.72	No	13.39	578	2.15	No	21.2	Rebar	Not Sustainable
1.17	No	23.4	1298	1.89	No	9.45	Rejar	Not Sustainable
2.59	No	49.02	6260	5.48	No	9.98	Retoy	Not Sustainable
1.81	No	37.41	552	11.5	No	16.3	Retoy	Not Sustainable
2.92	No	141.08	114	11.51	No	39.21	Rebottle	Not Sustainable
2.5	No	34.69	990	15.27	No	43.71	Ecopack	Not Sustainable
2.06	No	49.12	7052	1.75	No	18.45	Biobottle	Not Sustainable
2.3	No	46.81	90	15.1	No	10.93	Terrajar	Not Sustainable
2.36	No	155.17	724	24.45	No	7.6	Terrabar	Not Sustainable
2.82	No	5.68	271	12.35	No	30.93	Purejar	Not Sustainable
2.72	No	26.15	38	11.53	No	1.22	Biobar	Not Sustainable
2.46	No	5.08	64	1.63	No	8.96	Purebrush	Not Sustainable
2.25	No	48.45	3504	44.45	No	11.43	Greenpack	Not Sustainable
2.89	No	16.12	5286	2.39	No	15.65	Rebrush	Not Sustainable
3.21	No	18.59	567	1.94	No	11.86	Rebrush	Not Sustainable
2.95	No	11.65	1540	1.46	No	48.63	Biojar	Not Sustainable
2.84	No	162.14	521	10.22	No	14.28	Rejar	Not Sustainable
2.11	No	35.92	8539	9.22	No	15.24	Ecotowel	Not Sustainable
1.67	No	36.47	518	1.43	No	17.76	Retowel	Not Sustainable
2.81	No	10.26	559	7.93	No	45.55	Ecotoy	Not Sustainable
2.93	No	29.6	332	4.12	No	35.79	Bioshirt	Not Sustainable
1	No	71.75	197	1.2	No	37.48	Biocup	Not Sustainable
2.63	No	52.13	1974	11.58	No	25.07	Pureshirt	Not Sustainable
2.59	No	6.91	890	2.91	No	35.42	Retowel	Not Sustainable
2.88	No	18.4	477	5.8	No	25.27	Purecup	Not Sustainable
2.35	No	24.94	260	0.78	No	42.49	Terrabar	Not Sustainable



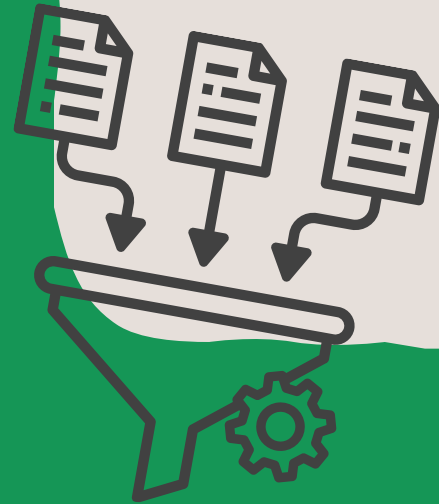
DATASET OVERVIEW



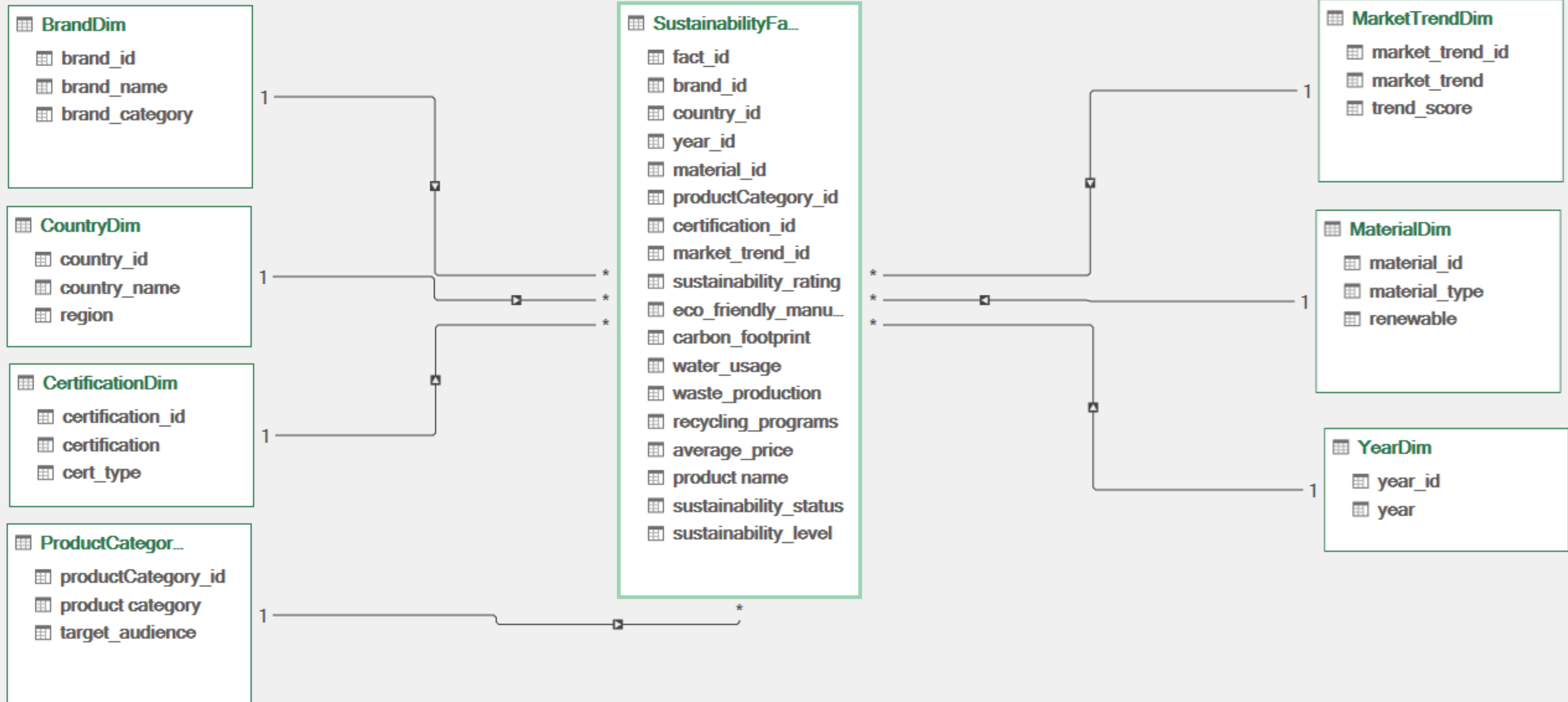
- Sustainability Project Data (CSV)
- From Kaggle
- Time Range: from 2010 to 2024
- The data consists of 5001 row & 15 column



EXCEL



THE MODELLING



FIRST DASHBOARD

Home

Dashboard 2

Average
Price 30.56



Avg. Carbon
Footprint 34.95



Avg. Water
Usage 833.93



Avg. Waste
Prod 6.71



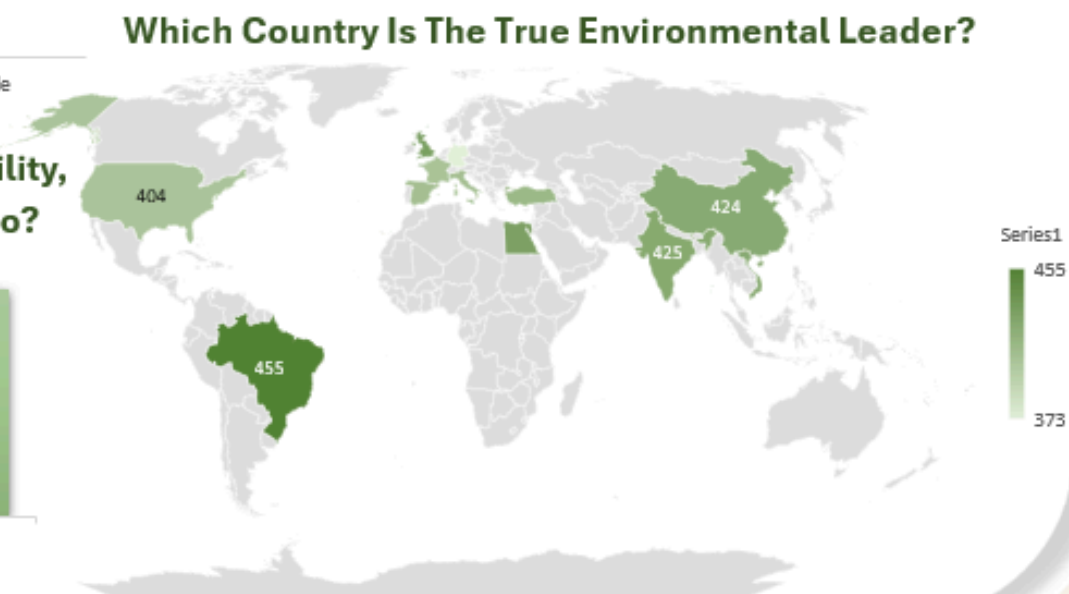
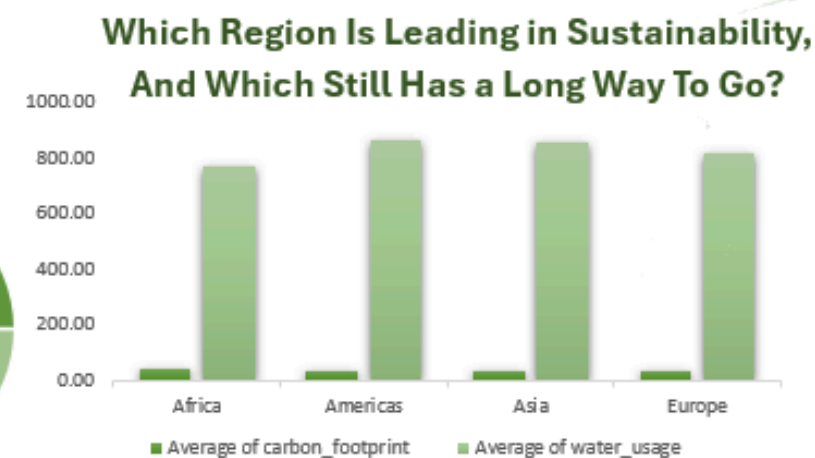
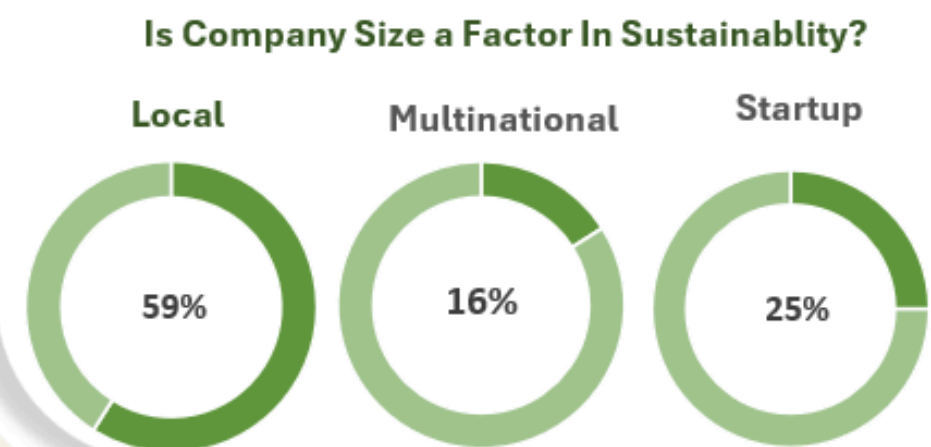
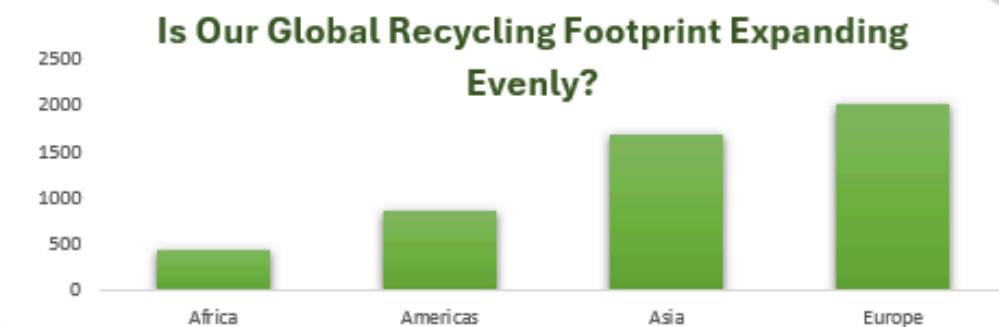
Min. Sus.
Rating 1



region	
Africa	Americas
Asia	Europe

country_name	
Brazil	China
Egypt	France
Germany	India
Italy	Spain
Turkey	Uk
Usa	Vietnam

year	
2015	2016
2017	2018
2019	2020
2021	2022
2023	2024
2025	



Home

Dashboard 1

market_trend

Declining Rising Stable

product category

Clothing Electronics Access...
Garden Home
Kitchenware Office Supplies
Packaging Personal Care
Toys

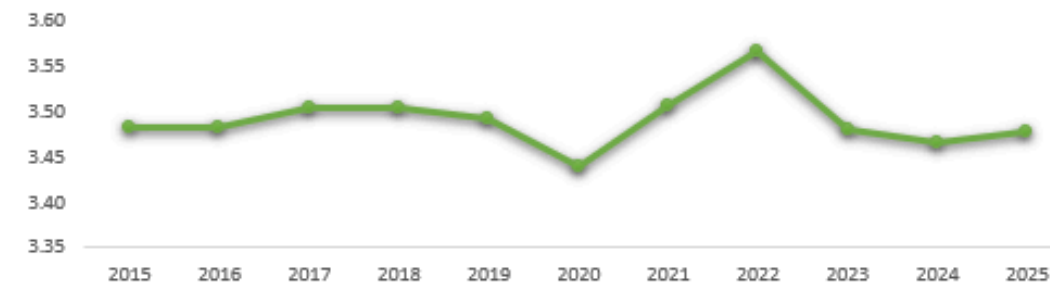
sustainability_status

Fully Sustainable
Not Sustainable
Partially Sustainable

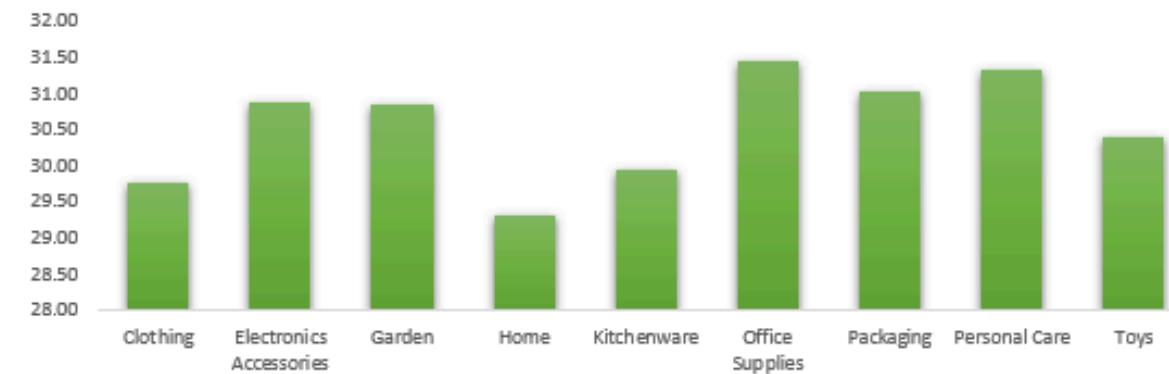
cert_type

Energy Environmental
Ethical None

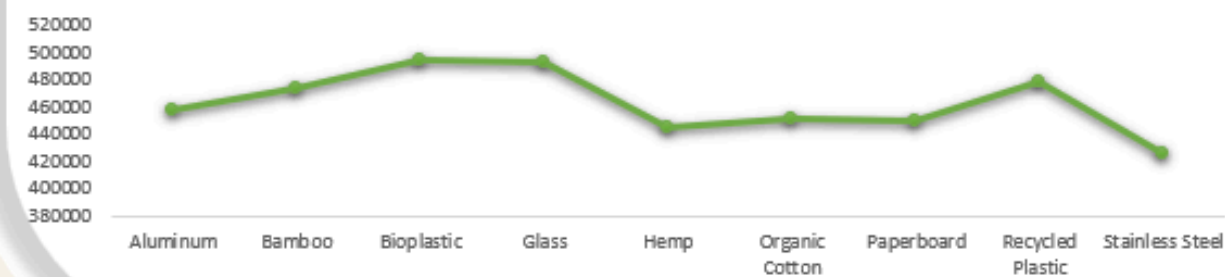
Are We On a Sustainable Trajectory?



Which Product Category Offers The Best Value?

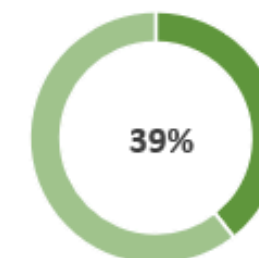


Which Materials Are The Most Water-Intensive?

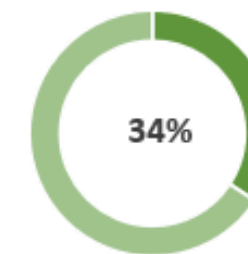


Does Sustainability Cost a Lot?

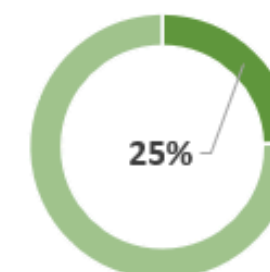
Fully Sustainable



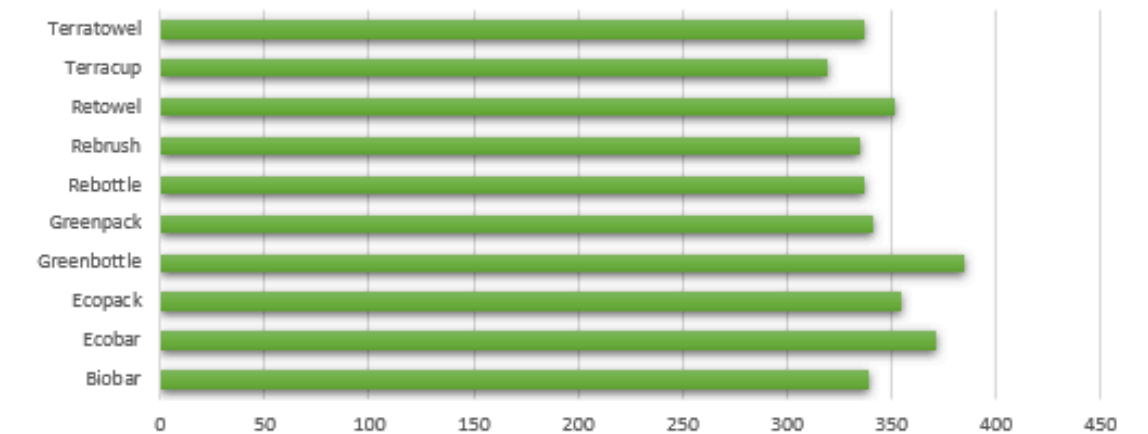
Partially Sustainable



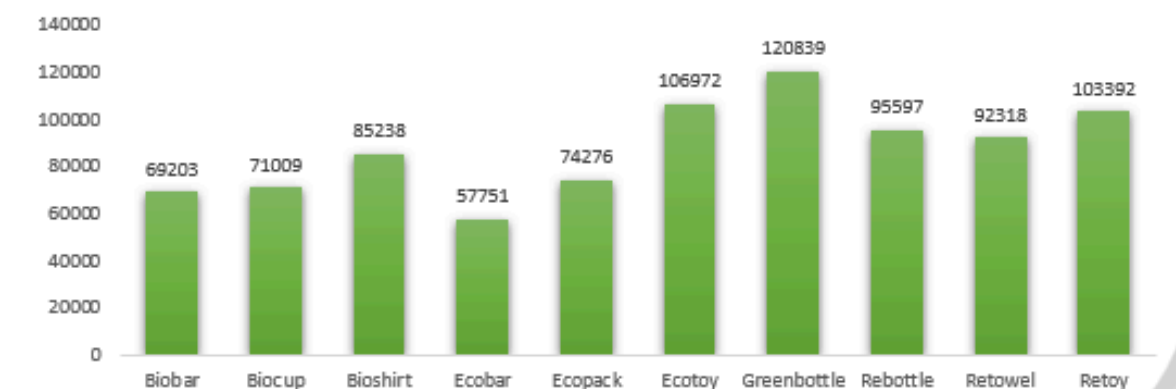
Not Sustainable



Who Are The Top 10 True Sustainability Leaders?



Which Of Our Top 10 Products Are Draining Our Water Resources?

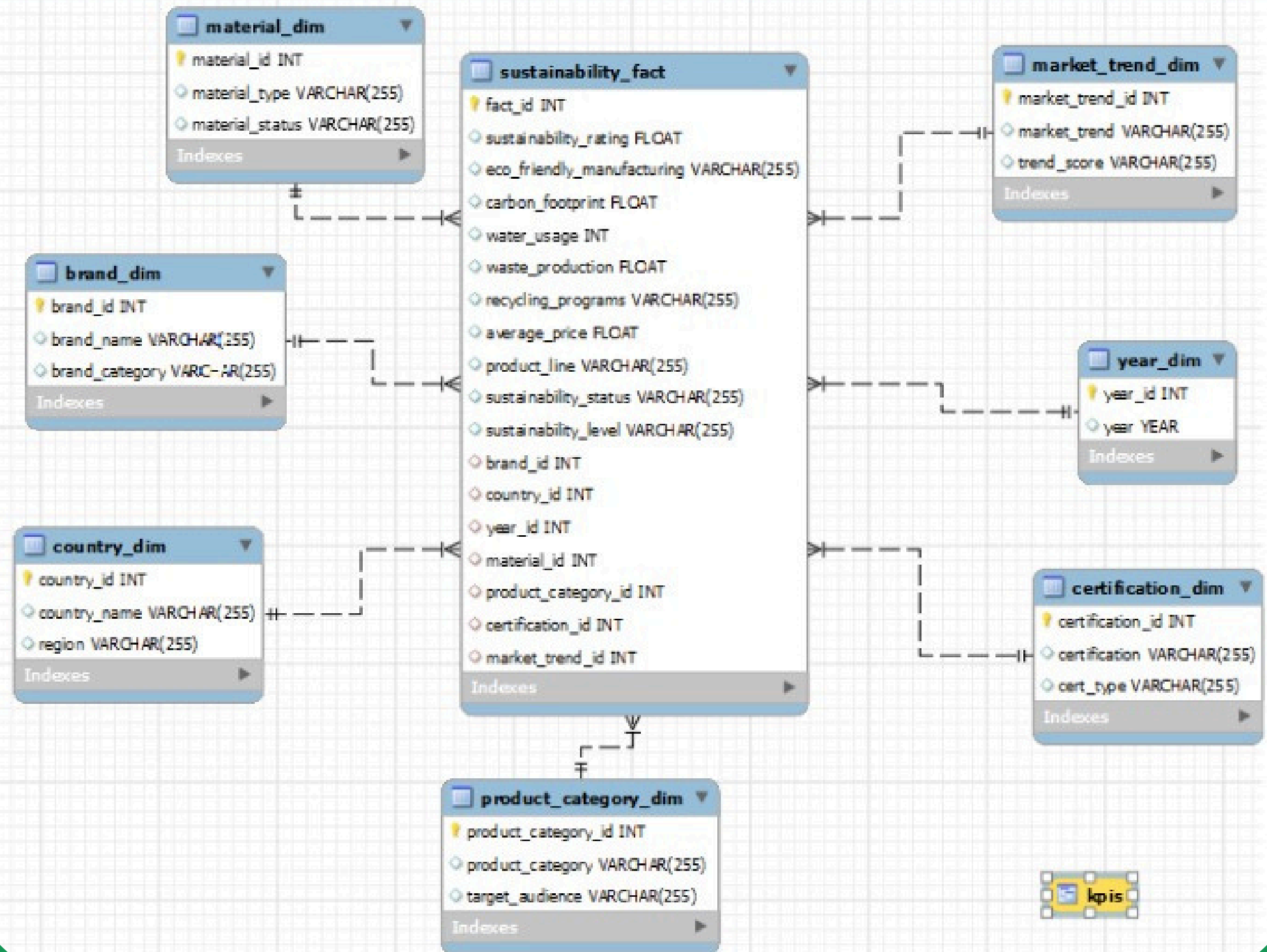
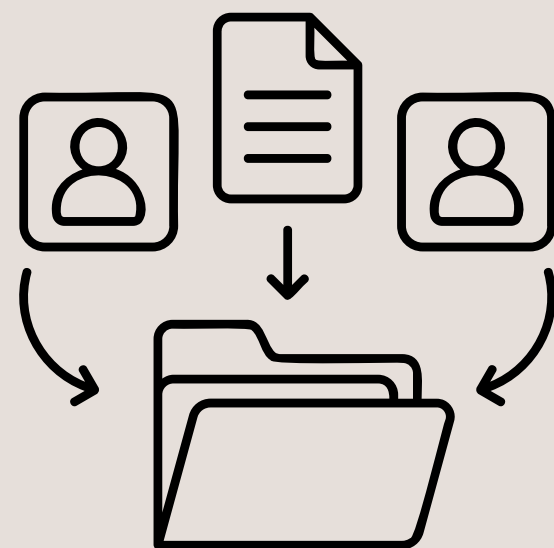


SECOND DASHBOARD

SQL

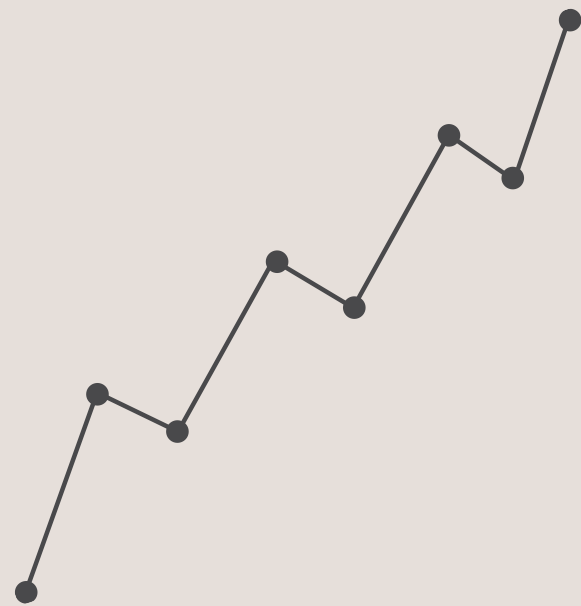


MODELLING

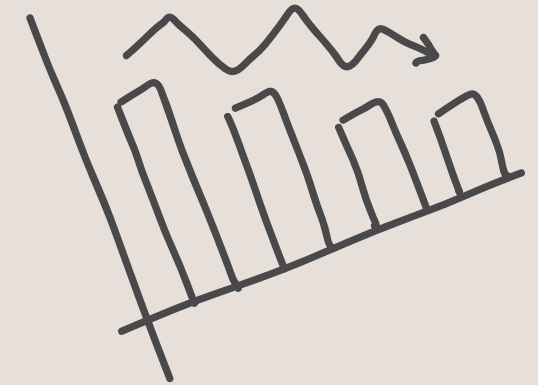


PYTHON





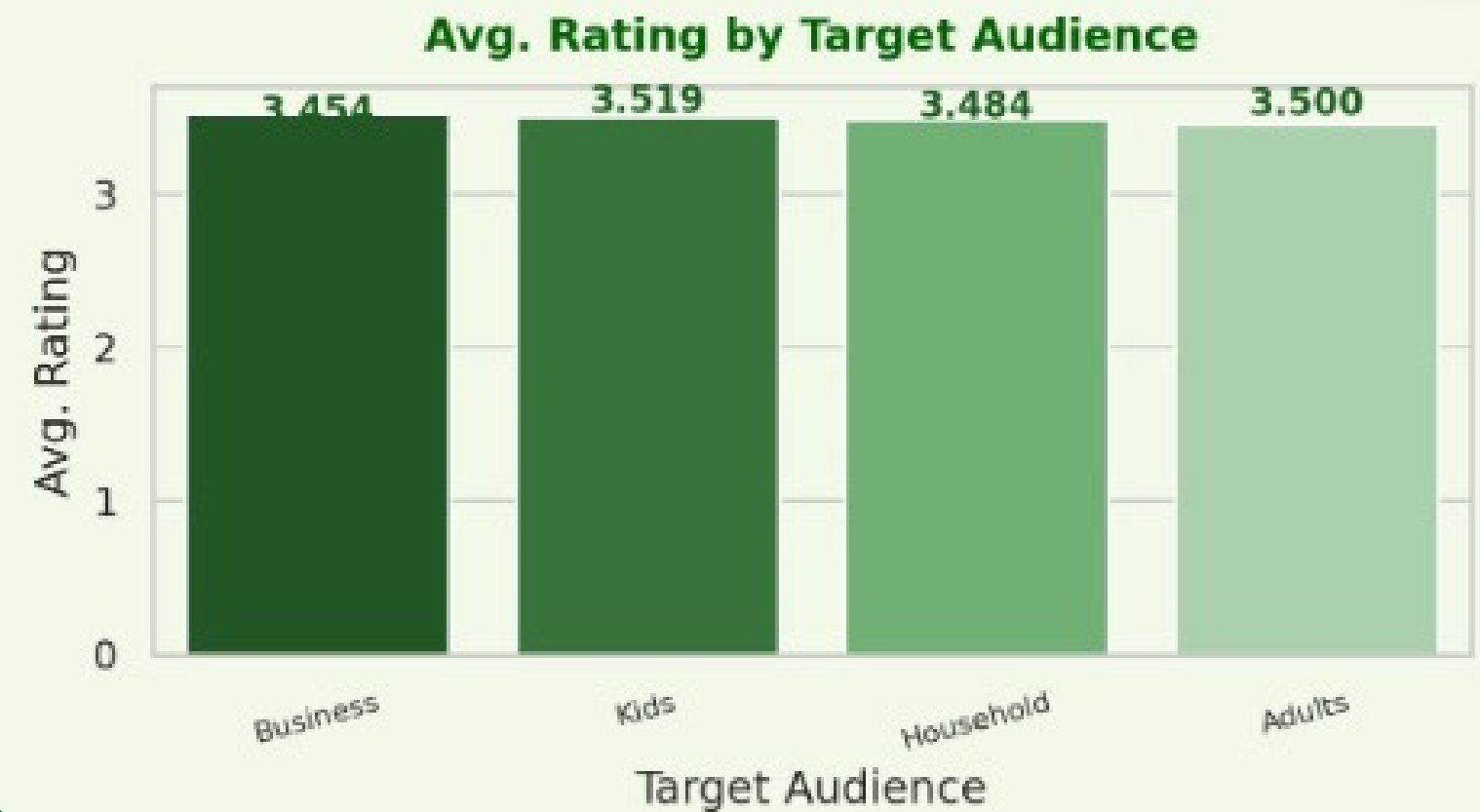
SOME GRAPHS



Avg. Rating Improvement Over Time

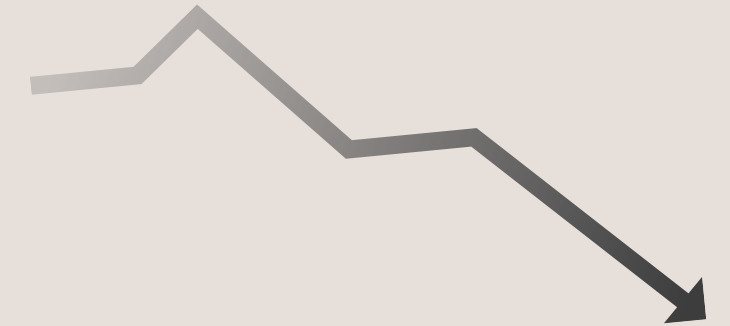


Avg. Rating by Target Audience





SOME GRAPHS



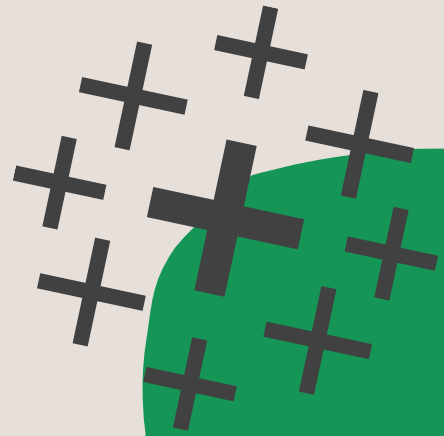
Price vs. Sustainability Rating



Impact of Certification on Rating



TABLEAU



FIRST DASHBOARD

Home

Dash 2

Dash 3

Dash 4

Dash 5

AVG Carbon
footprint

29



AVG price

32



AVG
Water
Usage

648



AVG Waste
Production

1,678



Max Sus
Rating

5



Min Sus
Rating

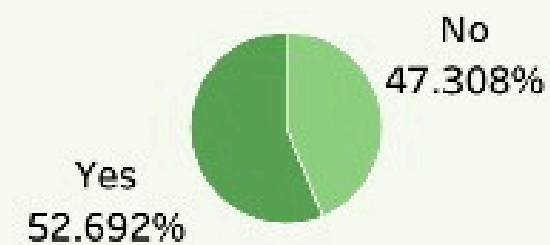
1



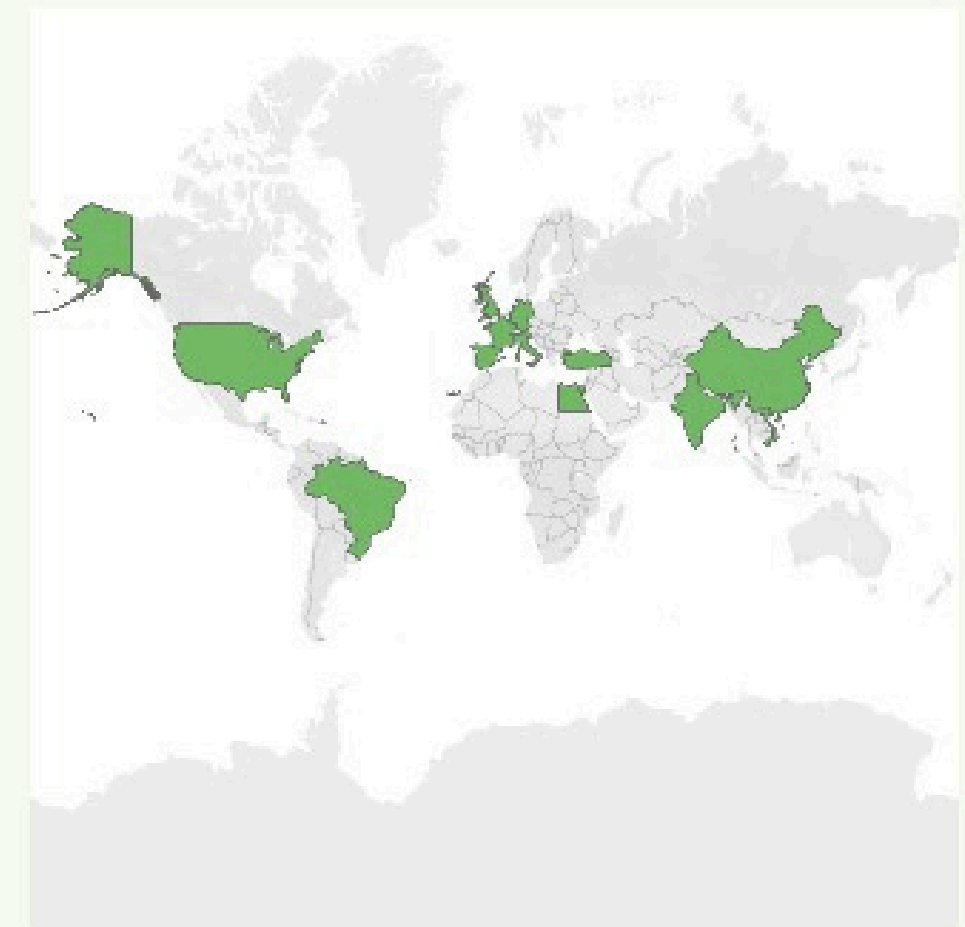
Environmental impact for top 5 sustainability rate



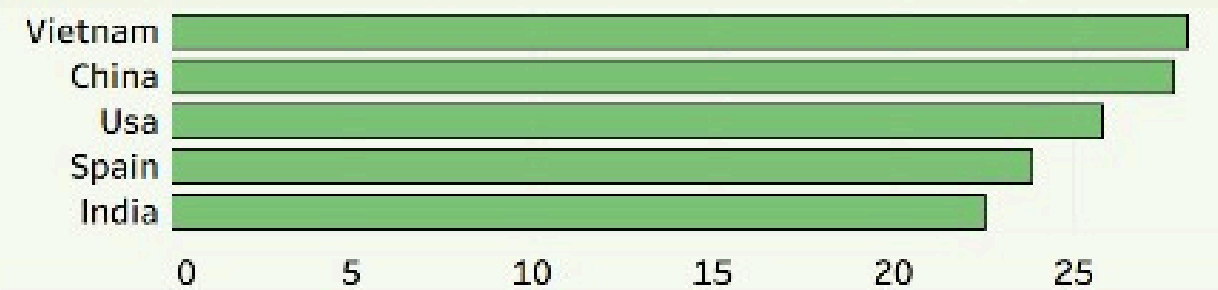
Sustainability rate for eco-friendly and non-eco-friendly



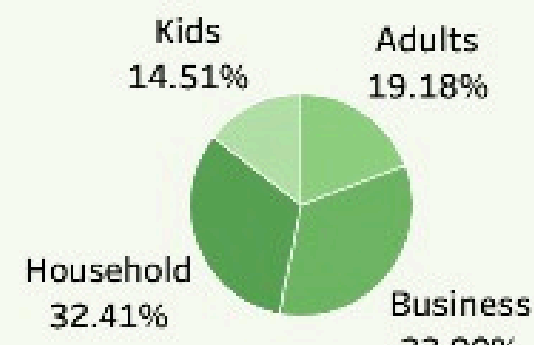
Certification by country



The lowest 5 countries in AVG carbon footprint



Carbon footprint for every target audience



Home

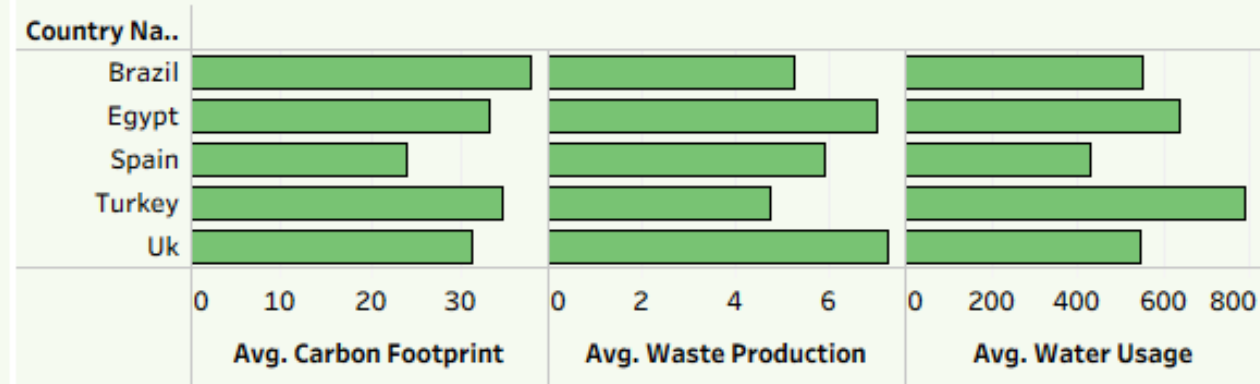
Das 1

Dash 3

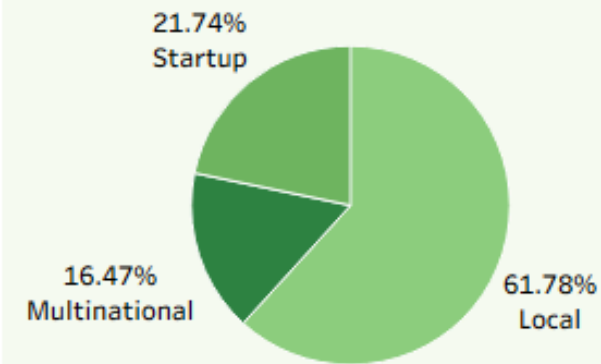
Dash 4

Dash 5

Environmental impact of the top 5 countries in sustainability rating



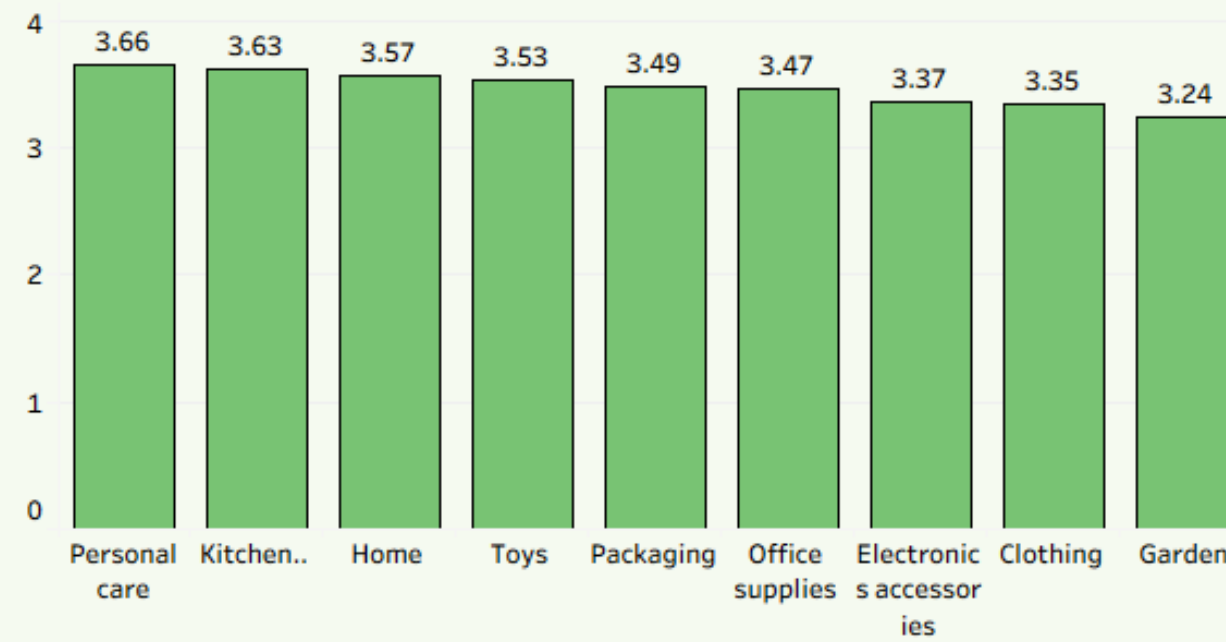
Avg price for every brand category



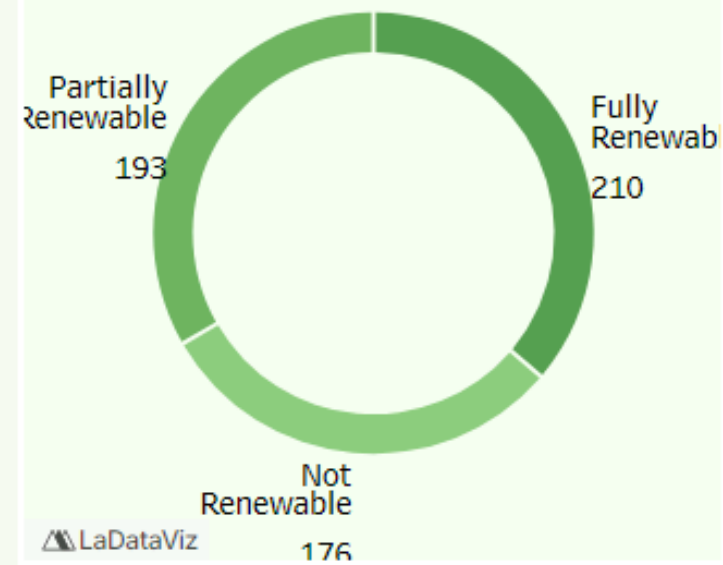
Environmental impacts over the years



Product line by sustainability rating



Distribution of material renewability



SECOND DASHBOARD

THIRD DASHBOARD

Home

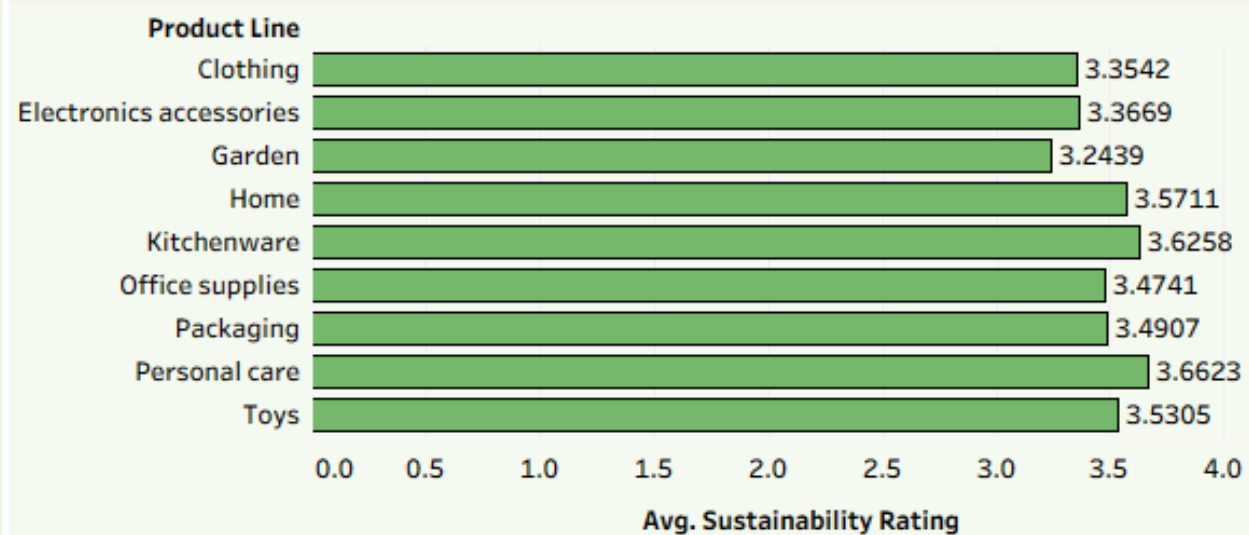
Dash 1

Dash 2

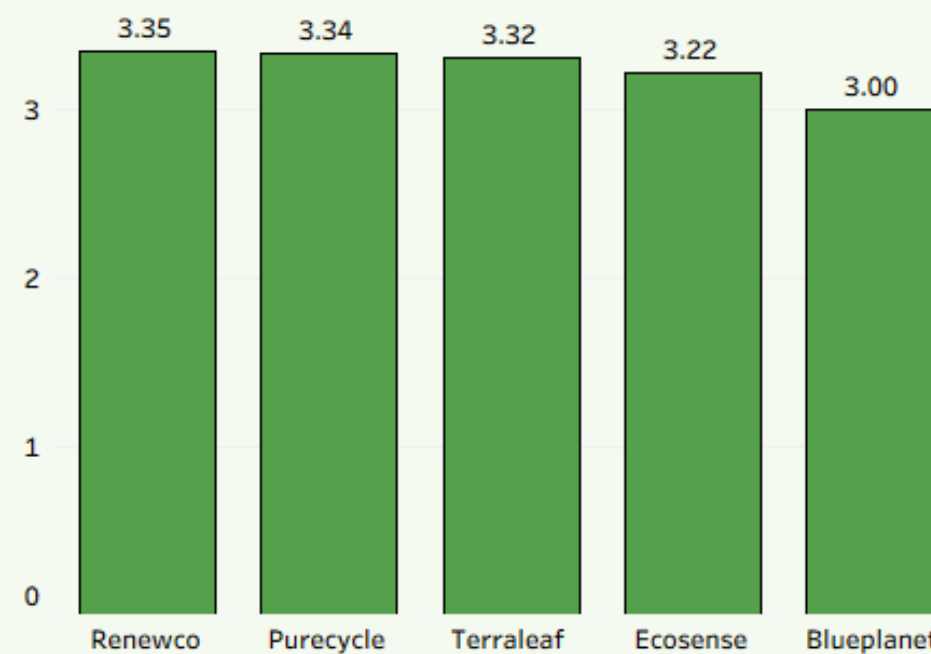
Dash 4

Dash 5

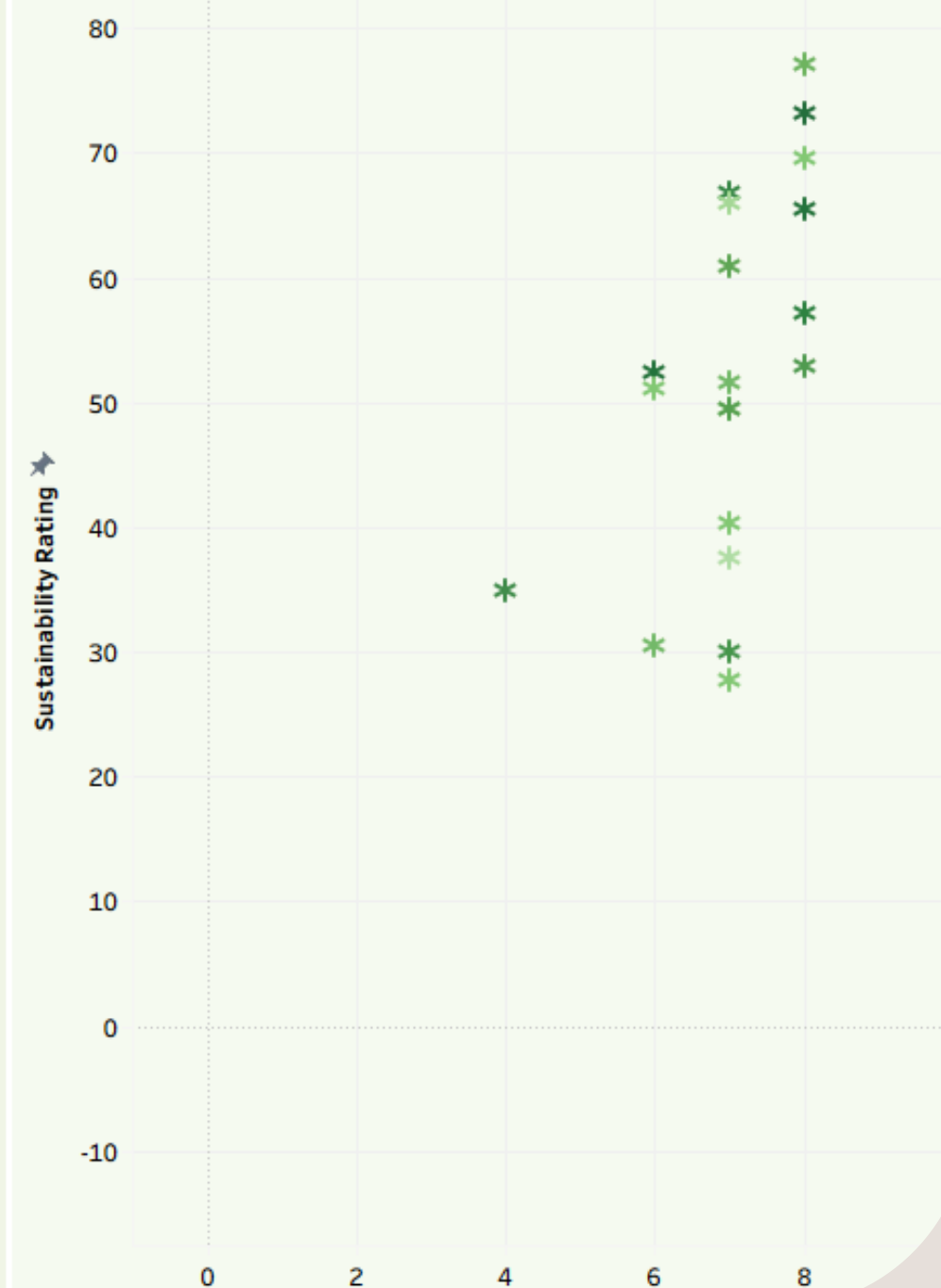
Environmental impact of each product category



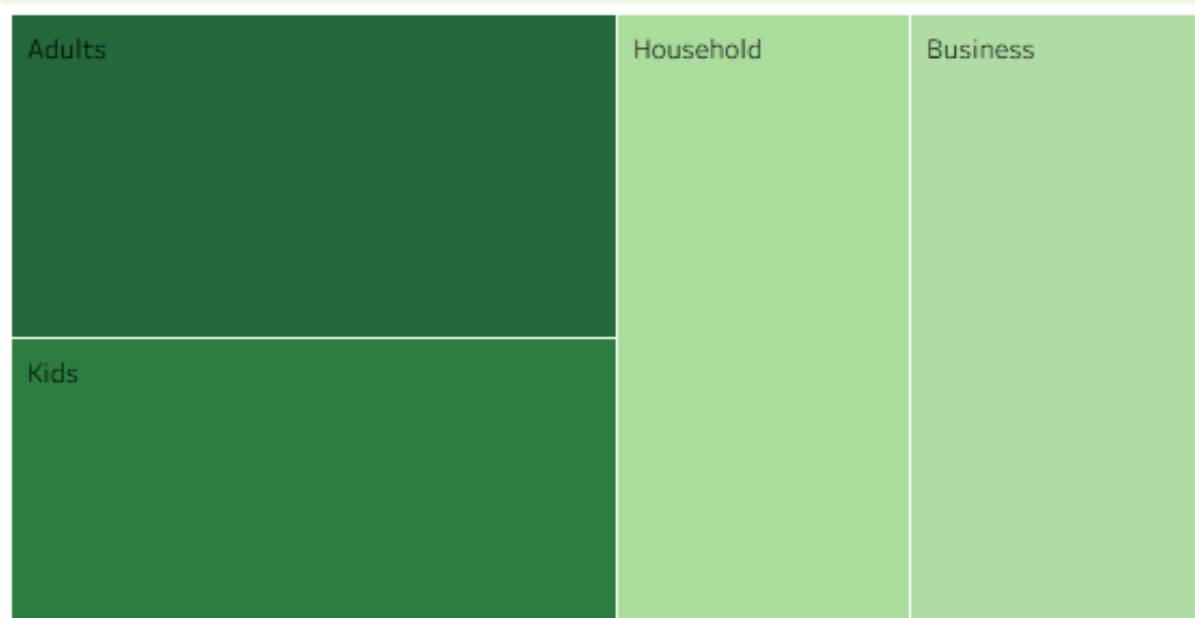
The bottom 5 brands in AVG sustainability rate



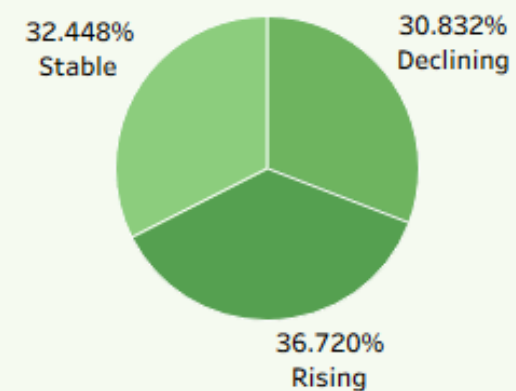
Impact of certification on sustainability



AVG Sustainability by target audience



Distribution of AVG Sustainability Rating by Brand Category



Home

Dash 1

Dash 2

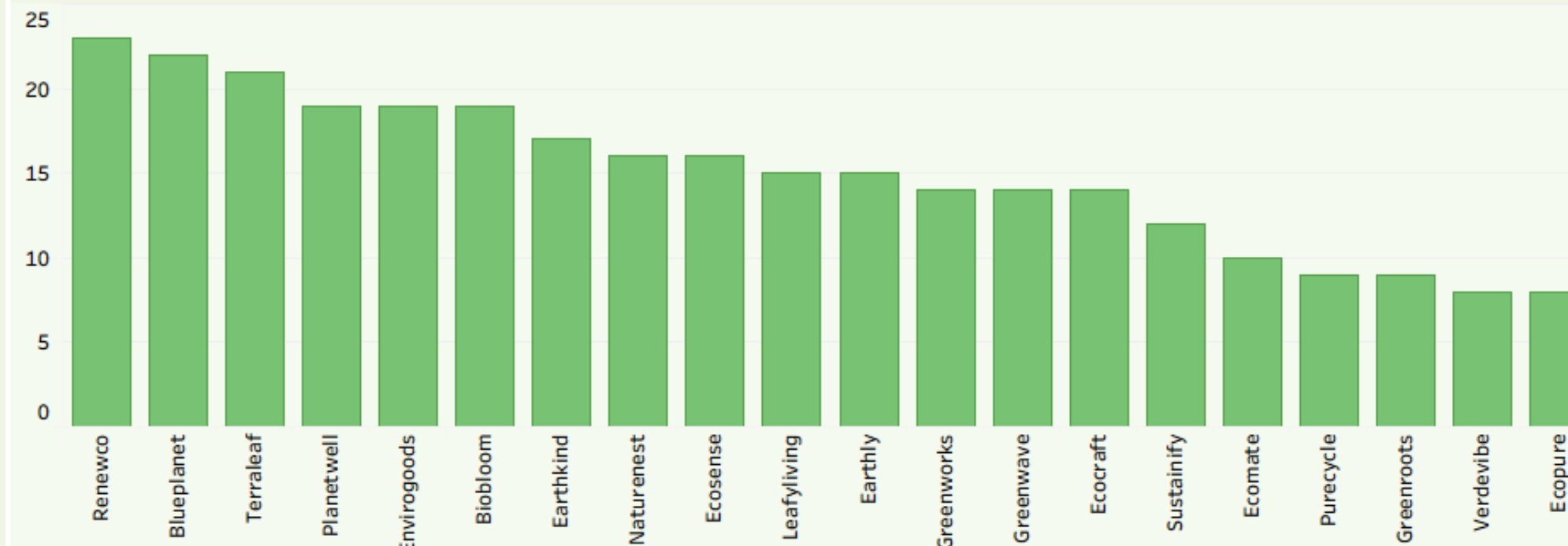
Dash 3

Dash 5

Average price by product line



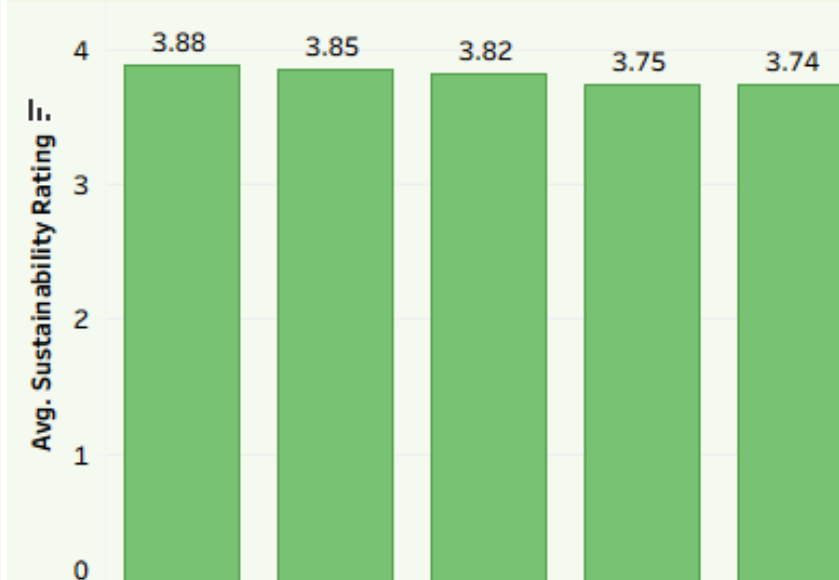
Brands with Recycling programs



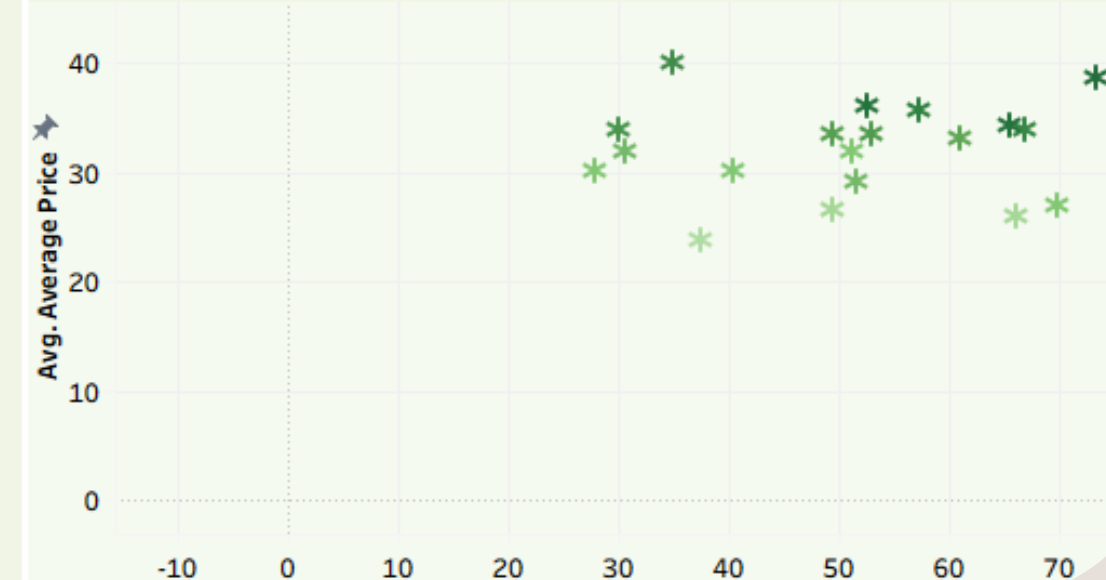
Sustainability for every Material Type



Top 5 sustainable brands



Correlation between AVG price and AVG Sustainability



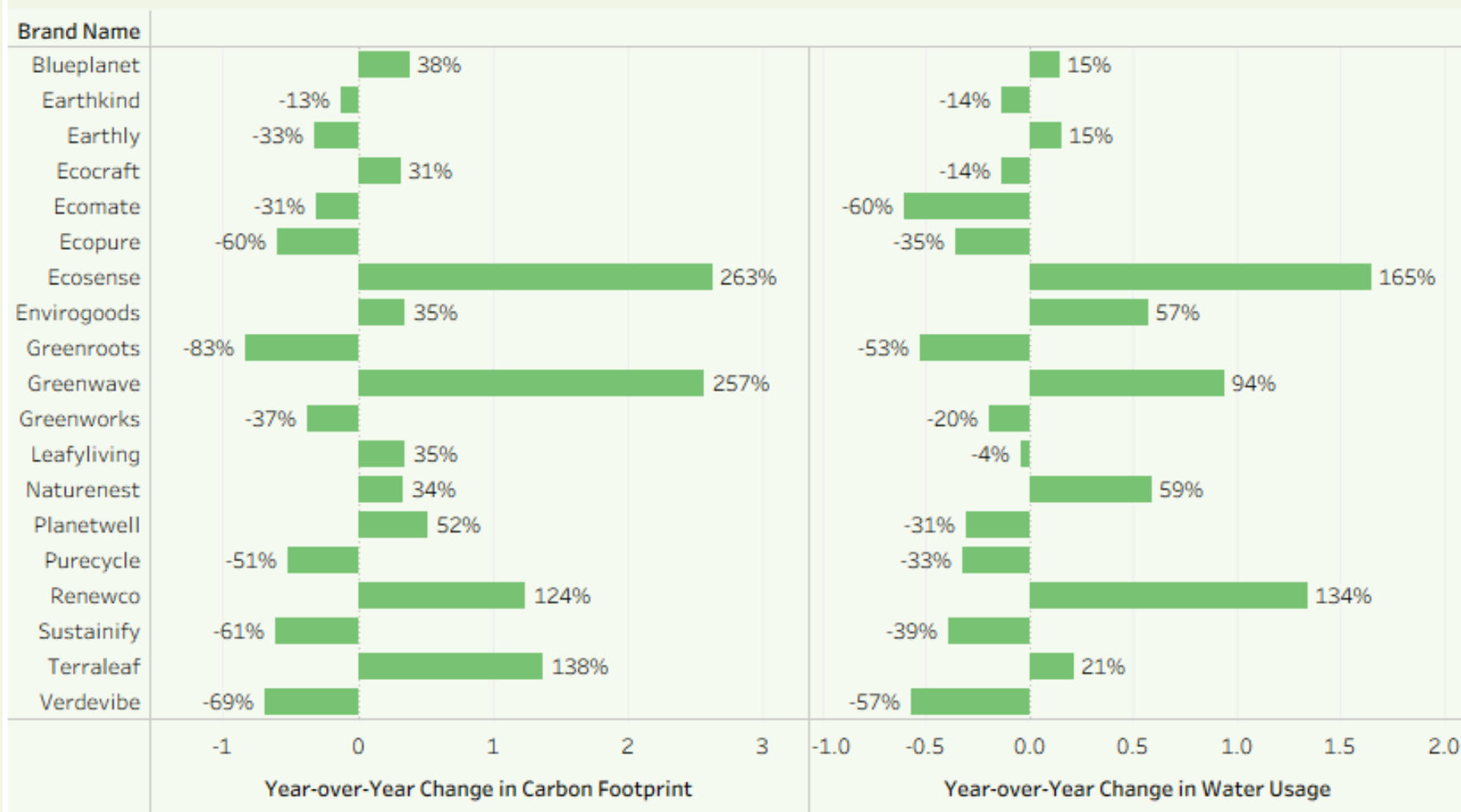
FOURTH DASHBOARD

FIFTH DASHBOARD



Sustainability Dashboard

Year Over Year Change for brands

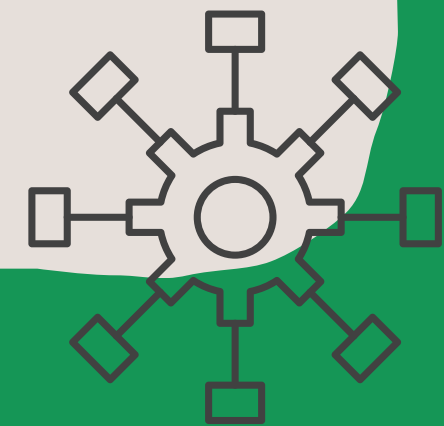


Waste ,Water ,Carbon efficiency per product line

Product Line	Carbon Efficiency	Waste Efficiency	Water Efficiency
Clothing	96%	25%	12%
Electronics accessories	113%	17%	14%
Garden	66%	16%	6%
Home	78%	18%	7%
Kitchenware	125%	27%	9%
Office supplies	123%	22%	11%
Packaging	127%	17%	10%
Personal care	63%	15%	8%
Toys	71%	16%	7%



POWER BI



FIRST DASHBOARD



Sustainability project

AVG Waste Production



6.23



AVE Water Usage

654



AVG Carbon FootPrint

30

Filters

country_n... ▾

All ▾

product_li... ▾

All ▾

year ▾

2015

2025

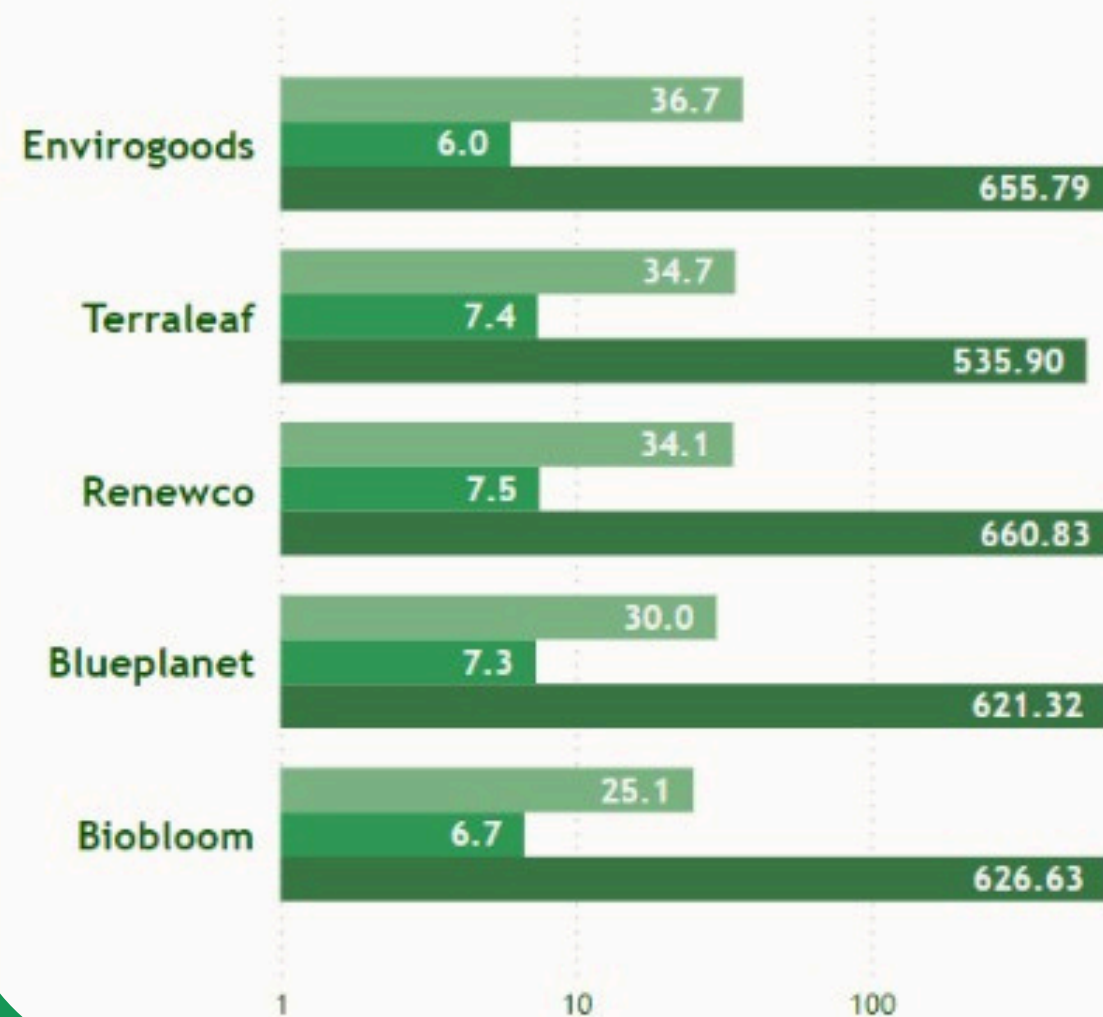
Home

Dashboard 2

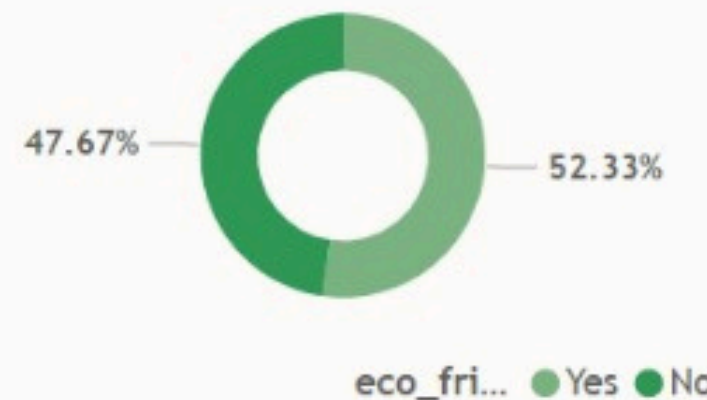
Dashboard 3

Dashboard 4

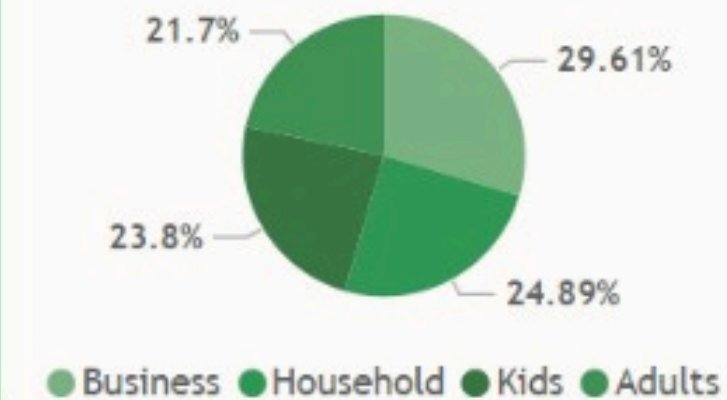
Enviromental impact for top 5 brands by sustainability rating



Sustainability rate for eco-friendly and non-eco-friendly



Carbon footprint for every target audience



AVE Carbon footprint for each country



Min Sust Rating



1

Max Sust Rating



5

AVG Price



32

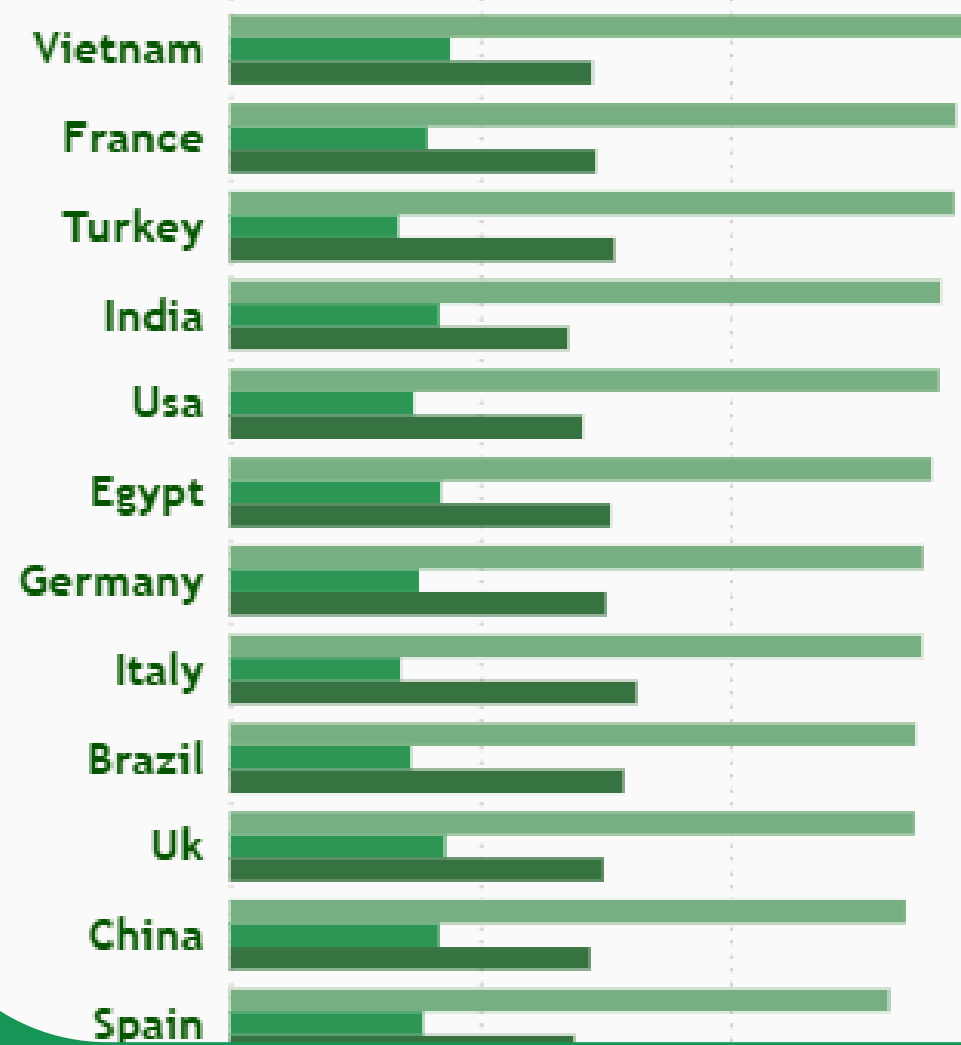
Home

Dashboard

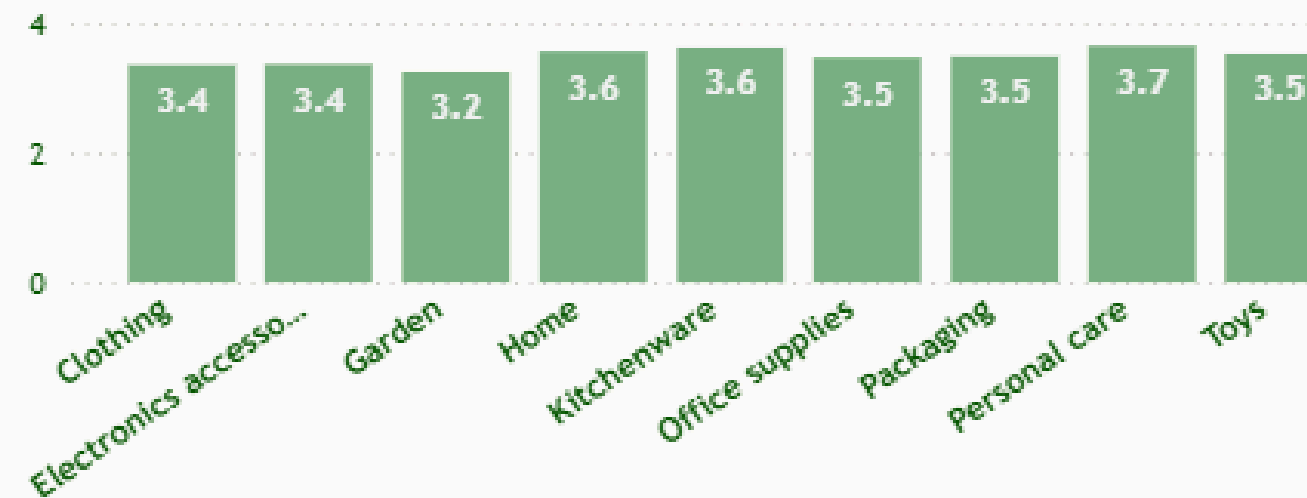
Dashboard 3

Dashboard4

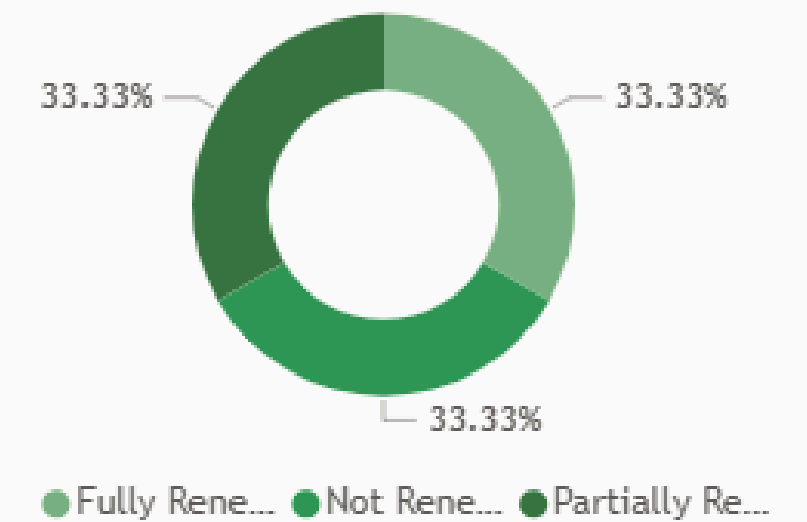
Enviromental impact for countries by sustainability rating



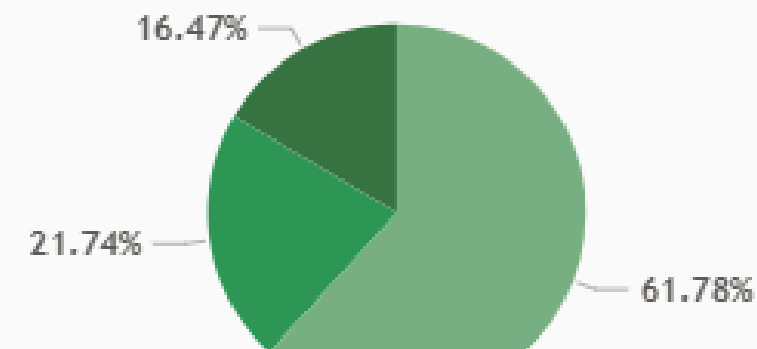
Product line by sustainability rating



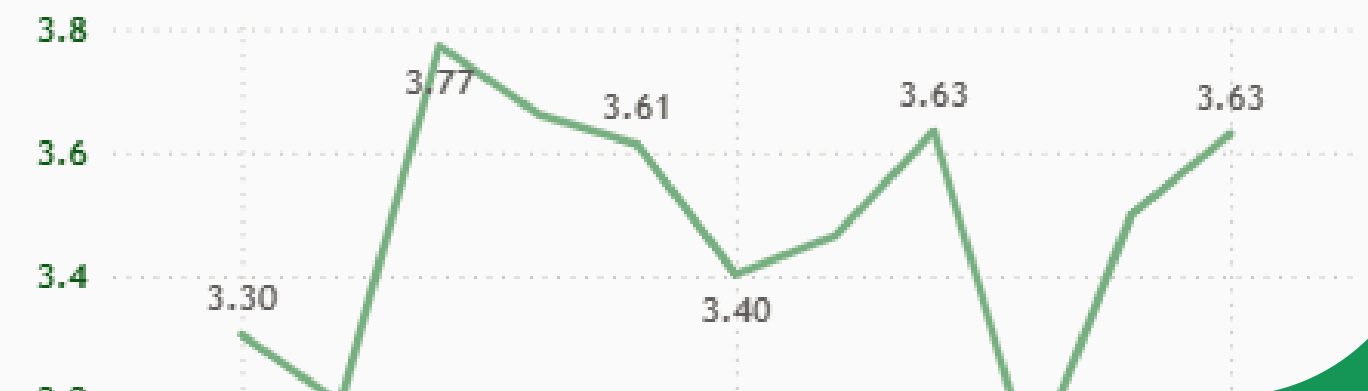
Distribution of material renewability



Avg price for every brand category

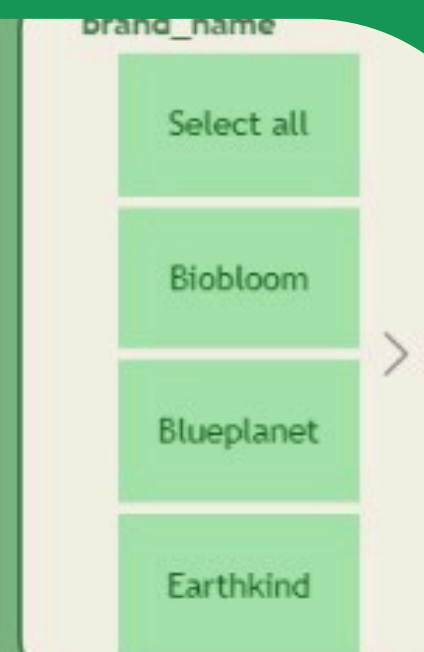
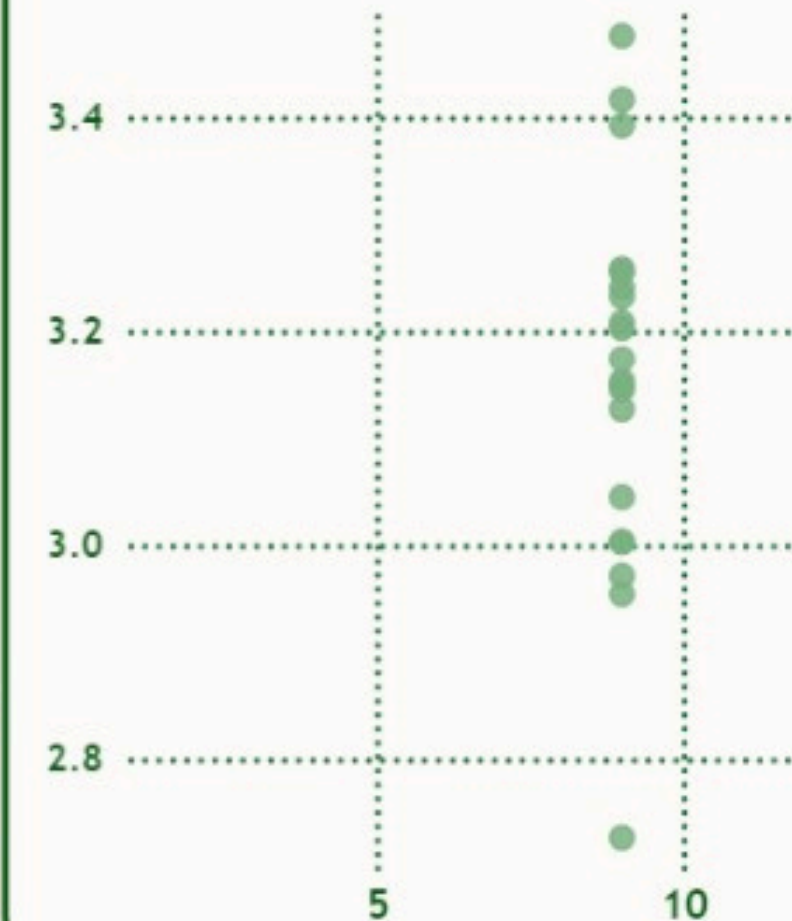
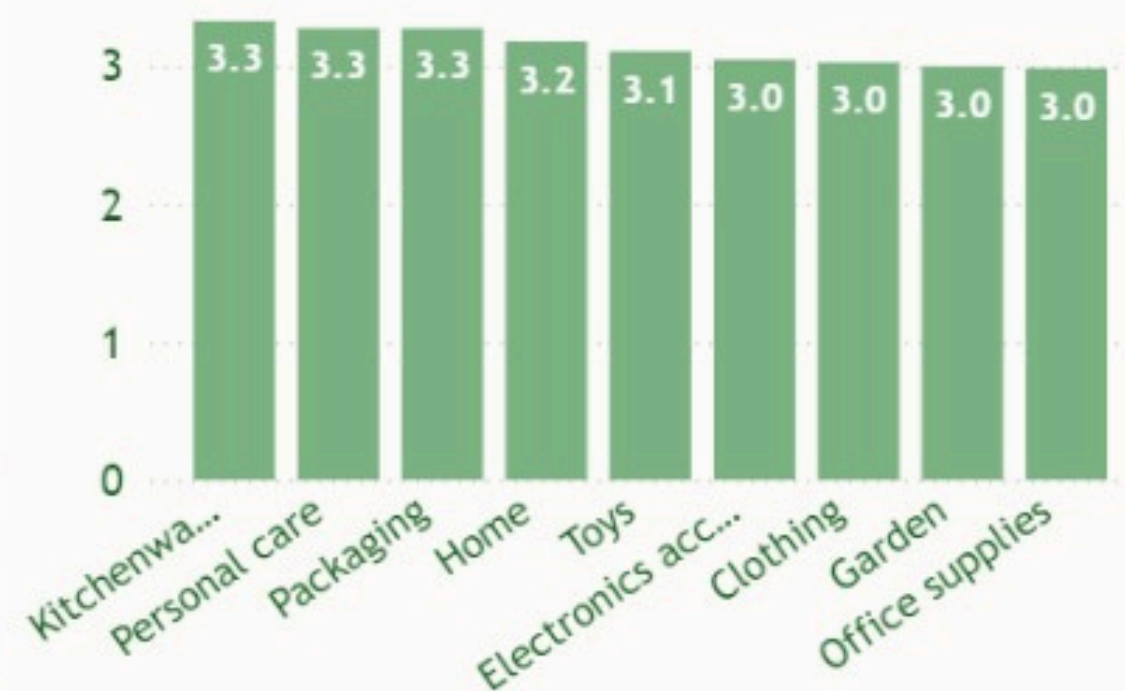
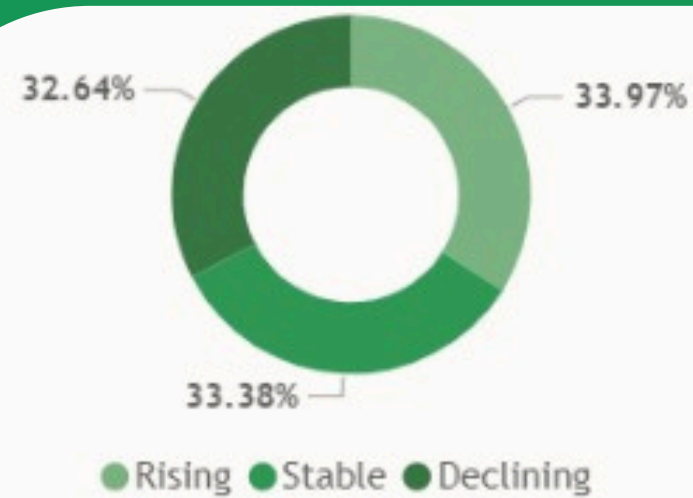


Average of sustainability_rating over year

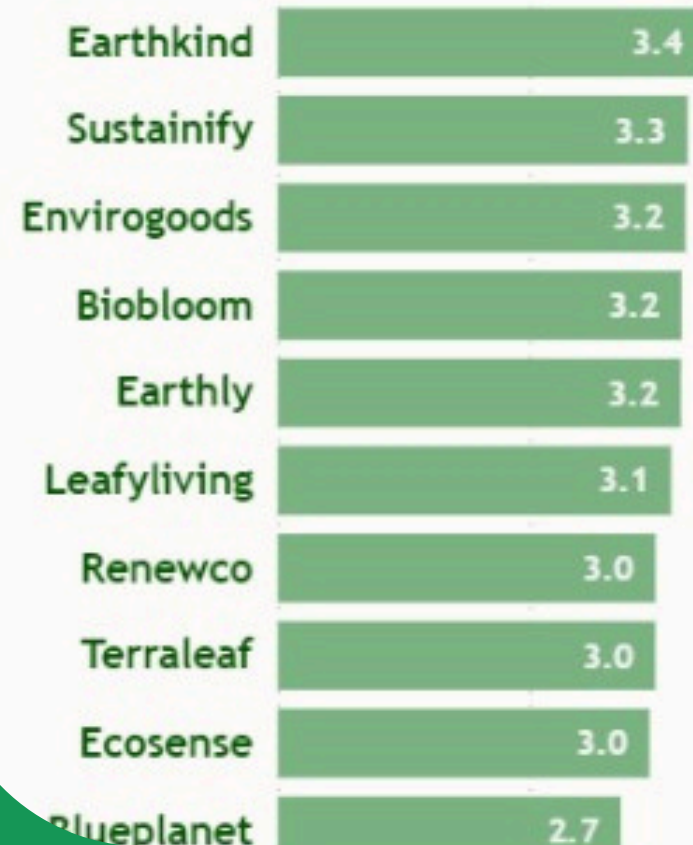


SECOND DASHBOARD

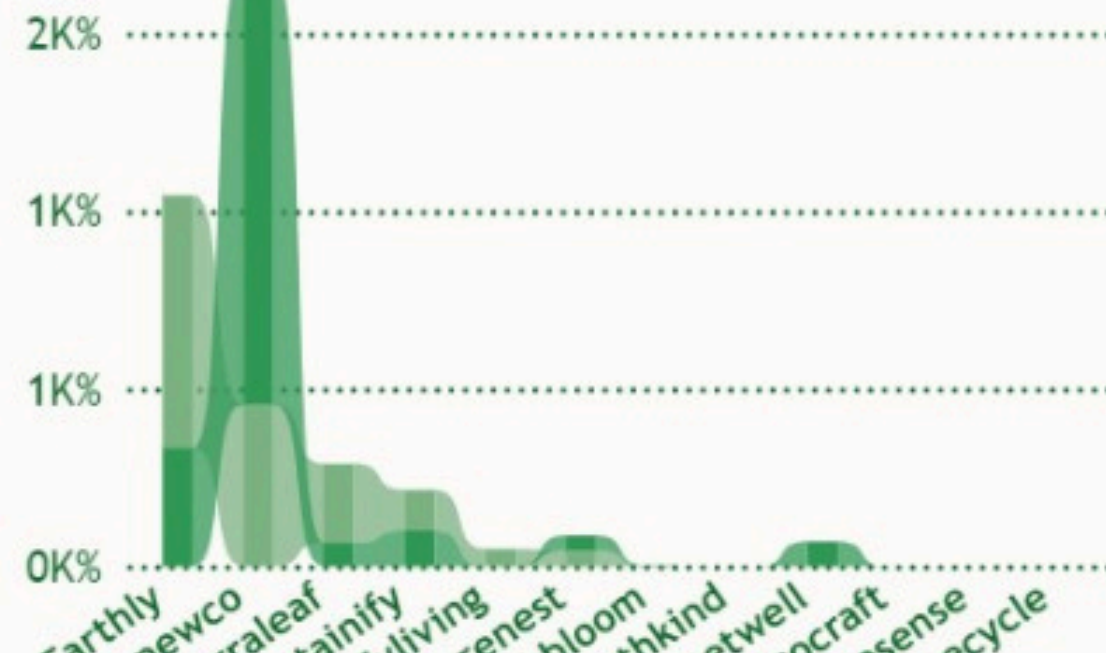
THIRD DASHBOARD



Average of sustainability_rating by brand_name



Year_over_Year_Change



AVE of sustainability_rating by target_audience



sustainabilit...

1.17

4.03

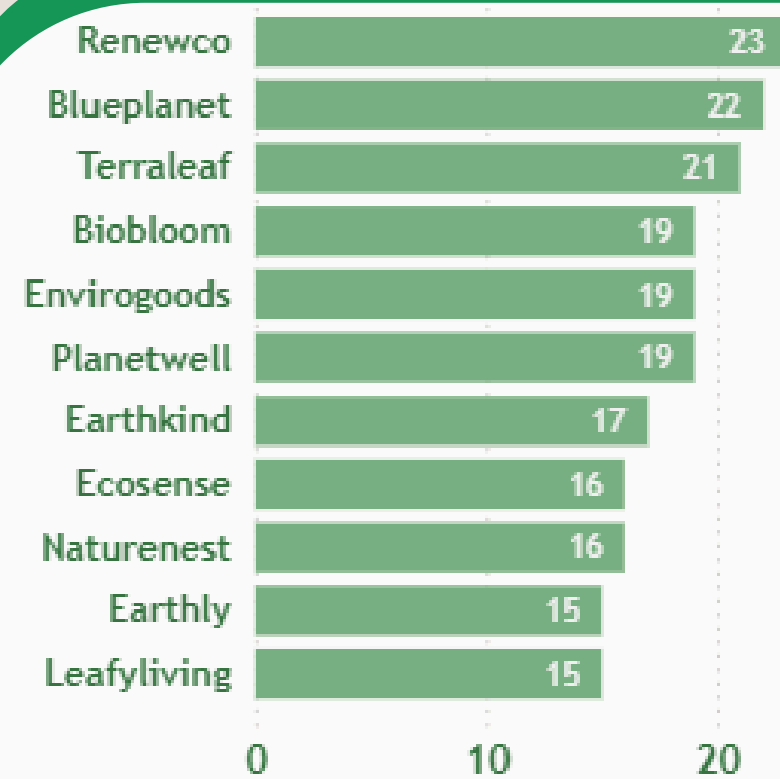


Home

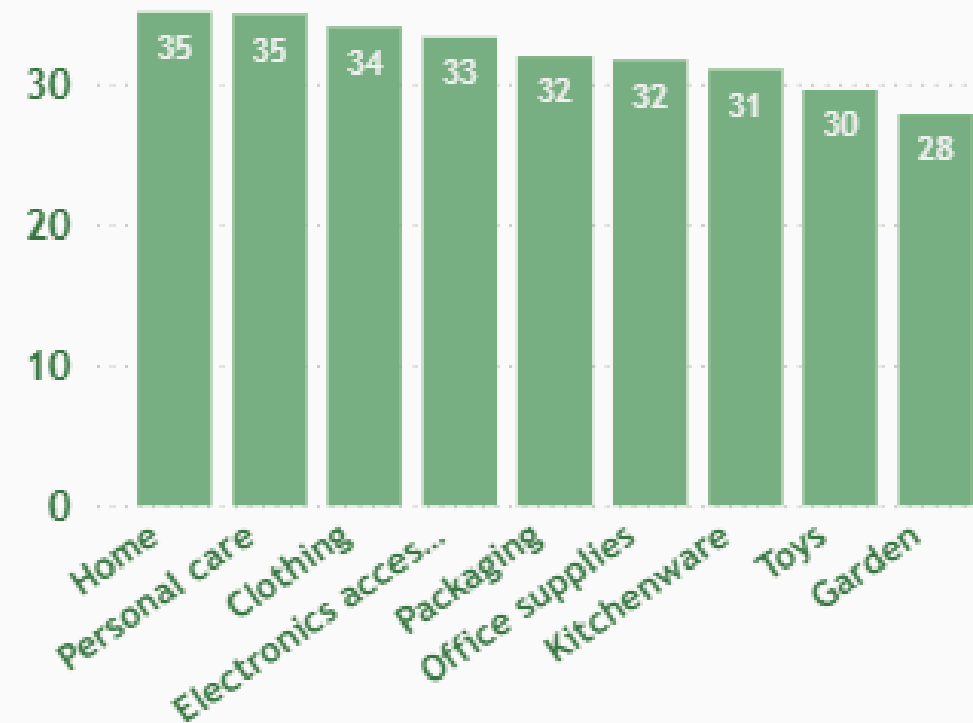
Dashboard 1

Dashboard 2

Dashboard 3



Clothing	25.73	0.25	0.96
	31.62	0.17	1.13
Garden	12.70	0.16	0.66
	16.10	0.18	0.78
Kitchenware	19.80	0.27	1.25
	23.35	0.22	1.23
Packaging	21.86	0.17	1.27
	17.32	0.15	0.63
Toys	16.27	0.16	0.71
Total	2.18	0.02	0.10



brand_name

Select all

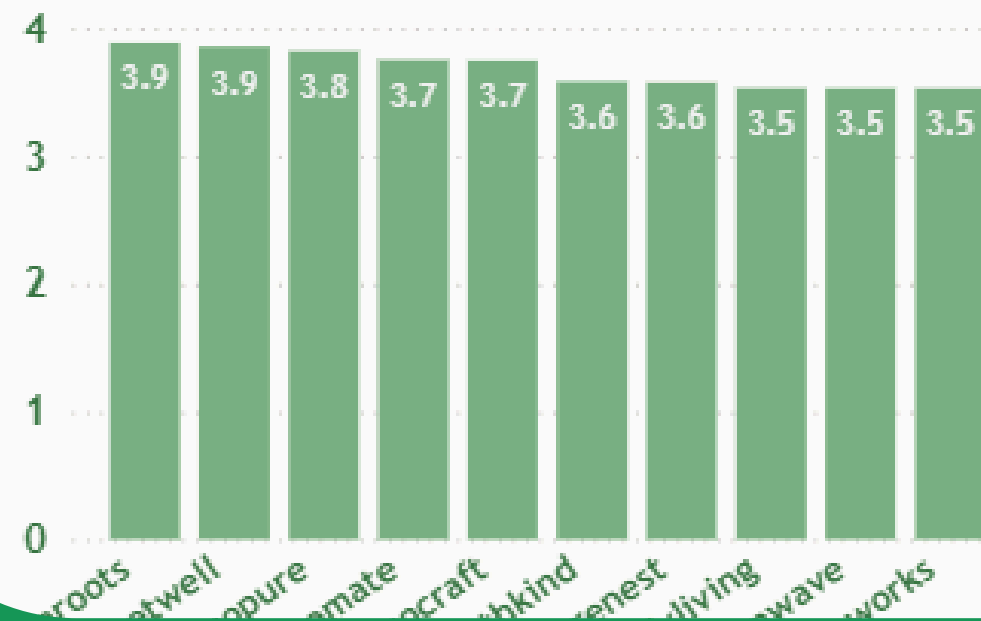
Biobloom

Blueplan...

Earthkind

Earthly

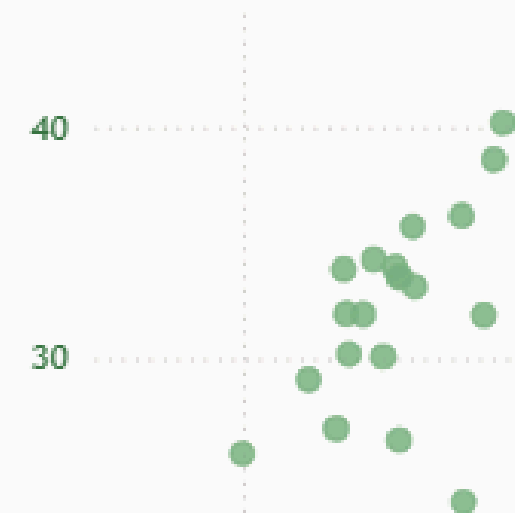
sustainability_rating by brand_name



sustainability_rating by material_type

Stainless steel	Hemp	Paperboard	Bioplastic
3.66			
Recycled plastic	3.49	3.48	3.43
3.62	Aluminum	Organic cotton	
Glass	3.33		
	Bamboo		

Correlation between AVG price and AVG Sustainability



Home

Dashboard 1

Dashboard 2

Dashboard 3

FOURTH DASHBOARD



**THANK YOU FOR
YOUR ATTENTION**

